Marcus Acosta

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EXPERIENCE

Bettorca **Remote**
Co-Founder & Engineer *Sep 2025 - Present*

- Trained probability models in **PyTorch/XGBoost** to forecast prediction market outcomes, achieving **0.687 ROC-AUC** across **270+** events.
- Engineered stateful point-in-time feature pipelines with temporal cutoffs to prevent data leakage across market and news streams.
- Processed **2.26M** historical observations via resumable data ingestion pipelines featuring event reconciliation and deduplication.
- Deployed a **Dockerized** ML scoring service on **Railway**, serving **34k+** requests with **37–136ms p95** latency and zero downtime.
- Integrated **Go/PostgreSQL** for asynchronous prediction logging, immutable state snapshots, and continuous model evaluation.
- Established MLOps workflows using **MLflow** for experiment tracking and **SHA-256** hash verification for artifact integrity.

OPEN SOURCE

DeepSpeed **GitHub**
Distributed Training Optimization

- Improved performance monitoring for distributed training loads by correcting **GPU bandwidth calculations**, eliminating a **50%** measurement overestimation.
- Identified unsafe cross-rank Mixture-of-Experts (MoE) timing aggregation and implemented safe distributed profiling to prevent deadlocks and misleading performance metrics.

Hugging Face TRL **GitHub**
Post-Training Reliability

- Added dataset size validation to post-training evaluation callbacks, preventing invalid sample counts and improving error handling with regression tests.
- Prevented DPO training crashes by validating reference model update intervals at initialization and rejecting zero or negative values.

SKILLS

Languages: Python, CUDA, Go, Bash, SQL

Frameworks & Libraries: PyTorch, XGBoost, scikit-learn, NumPy, SciPy, Hugging Face, DeepSpeed, FastAPI

Infrastructure: Docker, Kubernetes, ONNX, MLflow, GitHub Actions, Railway

Data: PostgreSQL, Qdrant, Redis, PyArrow, Apache Parquet

ADDITIONAL EXPERIENCE

LifeTime Inc. **Walnut Creek, CA**
Facility Operations Supervisor *Jan 2024 - June 2026*

- Led daily operations for a 5–7 person facilities team, coordinating schedules, vendors, inventory, safety inspections, and compliance documentation.

EDUCATION

Post-Baccalaureate Coursework in Computer Science **Oregon State University**

- Data Structures, Discrete Mathematics, Databases, Software Engineering, Web Development

Bachelor of Arts, Sociology **UC Santa Cruz**